

Optimal Block Time for AMM Liquidity Providers under Jump-Diffusion Prices

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Outline

- The case for faster blocks
- Model and the frictionless decomposition
- Fees, block time, and the jump floor
- Calibration and magnitudes
- The planner's block-time optimum
- What this means, and what it does not

The case for faster blocks

Where the argument for sub-second blocks comes from

Passive liquidity providers on an AMM lose to arbitrageurs whenever the external price moves — **loss-versus-rebalancing** (LVR), Milionis et al. (2022).

- Under geometric Brownian motion, a constant-product pool loses at the rate $\sigma^2 V/8$, independent of the block time.
- Add a swap fee γ and Poisson blocks of mean Δt (Milionis et al., 2023): the rate is multiplied by $F(\kappa)$, $\kappa = \gamma/(\sigma\sqrt{\Delta t})$ — the stationary probability that an arriving block offers a *profitable* trade.
- $F(\kappa) \rightarrow 0$ as $\Delta t \rightarrow 0$. Shorter blocks $\Rightarrow$ less LVR, all the way down.

The deployed prescription

“Shorten blocks.” Solana at 400 ms, Ethereum L2s at 2 s, app-chains chasing 50 ms — with LP welfare among the stated reasons.

The primitive that prescription assumes

The whole argument rests on the reference price being **continuous**. A diffusive gap has size $O(\sigma\sqrt{\Delta t})$, so halving the block halves what the arbitrageur finds.

Crypto prices are not continuous:

- Scaillet et al. (2020): frequent, clustered jumps in BTC at high frequency.
- Saef et al. (2024): BTC jumps on $\sim 58\%$, ETH on $\sim 32\%$ of trading days.
- My own ETH/USDT calibration: $\lambda = 283$ jumps/yr.

A jump does not care about Δt

A jump of size J opens a price gap of size J whatever the block schedule. Faster blocks clear each gap *sooner* — they do not make it *smaller*.

This talk in one line

Result

Under jump-diffusion the LVR rate splits into a diffusion channel the block schedule governs and a jump channel it cannot touch. For symmetric jump laws the second is an **exact lower bound** — a floor no block speed breaches.

What follows:

1. A zero-fee decomposition (Theorem 1).
2. With fees and blocks: a separated rate plus a two-sided bounded remainder (Theorem 2, Proposition 1).
3. The floor, exactly, and the $\sqrt{\Delta t}$ approach to it (Theorem 3, Corollary 1).
4. The LP-side planner's optimum: a cubic, invariant in pool size and in every jump parameter (Theorem 4).
5. ETH/USDT calibration: 471 bp/yr at 12 s against a floor of 125, optimum near 8 s.

Model and the frictionless decomposition

Setup

Milionis et al. with **one primitive changed**: the reference price jumps.

$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_t + (e^{J_t} - 1) dN_t, \quad N \text{ Poisson}(\lambda), \quad J \sim \nu.$$

- ν finite activity with $\int (e^{j/2} - 1)^2 \nu(dj) < \infty$; Merton $J \sim \mathcal{N}(m, \delta^2)$ the running instance.
- CPMM, $x_t y_t = L$, pool value $V(P) = 2\sqrt{LP}$ — concave, and that concavity is the loss.
- Fee γ in log-price units $\Rightarrow$ no-arbitrage band $\{|\log S - \log P| \leq \gamma\}$; blocks Poisson of rate $1/\Delta t$.
- $\text{LVR}(T) = \int_0^T V'(P_{s-}) dP_s - [V(P_T) - V(P_0)]$; rate $\ell = \mathbb{E}_t[d\text{LVR}_t]/dt$.

What is general and what is not

Results need **symmetry** of ν , not Gaussianity. Theorems 1, 2, 4 hold for arbitrary ν ; Proposition 1 and Theorem 3 need symmetry.

Theorem 1 — the zero-free decomposition

In the continuous-arbitrage limit ($\gamma = 0$, $\Delta t \rightarrow 0$), $P_t \equiv S_t$ and LVR is the Itô–Lévy remainder of $V(S_t)$:

$$\ell = \underbrace{\frac{\sigma^2}{8} V(S_t)}_{\text{diffusion-LVR}} + \underbrace{\frac{\lambda}{2} V(S_t) \mathbb{E}[(e^{J/2} - 1)^2]}_{\text{jump-LVR}},$$

and for Merton $\mathbb{E}[(e^{J/2} - 1)^2] = e^{m+\delta^2/2} - 2e^{m/2+\delta^2/8} + 1 \geq 0$, zero iff $J \equiv 0$.

Two observations

The jump term carries **no** Δt . At $\gamma = 0$ the arbitrageur clears any gap in the next block, so refining the schedule only redistributes *when* the loss lands.

Measure

The drift cancels between the two terms, so no normalisation of μ is needed — but with jumps the rate is *not* measure-independent, so this is a physical-measure statement and calibration is against realised prices.

Fees, block time, and the jump floor

The mispricing is a persistent state

Write $z_t := \log(S_t/P_t)$. At a block the arbitrageur trades iff $|z| > \gamma$ and moves the pool to the *near band edge*: $z \mapsto \text{bound}(z, -\gamma, \gamma)$ — not to 0.

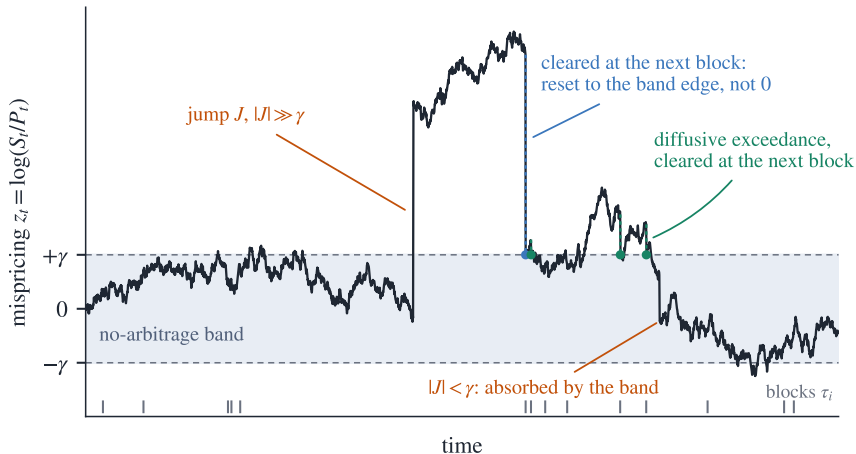
$$z_t = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t + \underbrace{\sum_{k: T_k \leq t} J_k}_{\text{exogenous, rate } \lambda} + \underbrace{\sum_{i: \tau_i \leq t} \mathcal{J}_i}_{\text{endogenous, rate } 1/\Delta t}$$

The asymmetry every result comes from

Two point processes drive z : price jumps push it **away** from the band at a rate the protocol does not choose; block-triggered resets push it **back** at a rate the protocol does choose.

Per block, loss $h(\xi) = \frac{\gamma}{2}(e^{\xi/2} - 1)^2$ with $\xi = \text{sign}(z)(|z| - \gamma)_+$, and $\ell(\Delta t) = \frac{1}{\Delta t} \mathbb{E}_\pi[h(\xi)]$ against the stationary law π .

A sample path of the mispricing z_t



Diffusive exceedances (teal) scale with $\sigma\sqrt{\Delta t}$; a jump (orange) opens a Δt -independent gap, cleared in full at the next block.

The two channels, separately

Diffusion (Lemma 1). At $\lambda = 0$ the stationary law is uniform on the band with exponential tails, so

$$\ell_{\text{diff}} = \frac{\sigma^2}{8} V \cdot F(\kappa), \quad F(\kappa) = \frac{1}{1 + \sqrt{2} \kappa}$$

with $\kappa = \gamma/(\sigma\sqrt{\Delta t})$; $F(\kappa) = \mathbb{P}_\pi(|z| > \gamma)$, strictly decreasing, $F(\kappa) \rightarrow 0$.

Jumps (Lemma 2). The jump contribution is $\lambda V G_\nu(\gamma)$ with

$$G_\nu(\gamma) = \frac{1}{8} \mathbb{E}_\nu[(|J| - \gamma)^2 \mathbf{1}\{|J| > \gamma\}],$$

a functional of ν alone. For symmetric Merton, $G = \frac{\delta^2}{8} \Psi(\gamma/\delta)$.

The fee discount is different on each channel

Both channels are trimmed by the fee — but by $F(\gamma/\sigma\sqrt{\Delta t})$ on one and by $\Psi(\gamma/\delta)$ on the other.

Only the first contains Δt , because the size of a jump is set by δ , not by how long the block lasted.

Theorem 2 — the separated rate

$$\ell(\Delta t) = \ell_0(\Delta t) + \mathcal{E}(\Delta t), \quad \ell_0(\Delta t) := \frac{\sigma^2}{8} V \cdot F\left(\frac{\gamma}{\sigma\sqrt{\Delta t}}\right) + \lambda V \cdot G_\nu(\gamma).$$

This is a *definition* of $\mathcal{E}$ and carries no content until $\mathcal{E}$ is controlled.

Proposition 1 bounds it on both sides — six signed terms, no order symbols, under a stated mixing condition $n_* \leq 2(1 + \eta^2)$ checked numerically over every block time, volatility and fee tier considered:

- interaction ($|M| \leq \gamma$ pathwise), block clock, single-jump thinning, multi-jump, curvature residual (exact, closed form), stationary law.
- At the calibration: $\mathcal{E}(\Delta t) \in [-0.653, +0.849]$ bp/yr at 12 s, never above 0.29% of the rate over 50 ms–12 s. Quadrature puts the realised remainder at +0.165 bp/yr, inside the bound and of the predicted sign.

Theorem 3 — an exact floor

No approximation, no remainder

Let ν be symmetric and satisfy $\mathbb{E}[g(\sum_1^k J)] \geq k \mathbb{E}[g(J)]$ for every k (Merton with $m = 0$ does).
Then for every $\Delta t > 0$

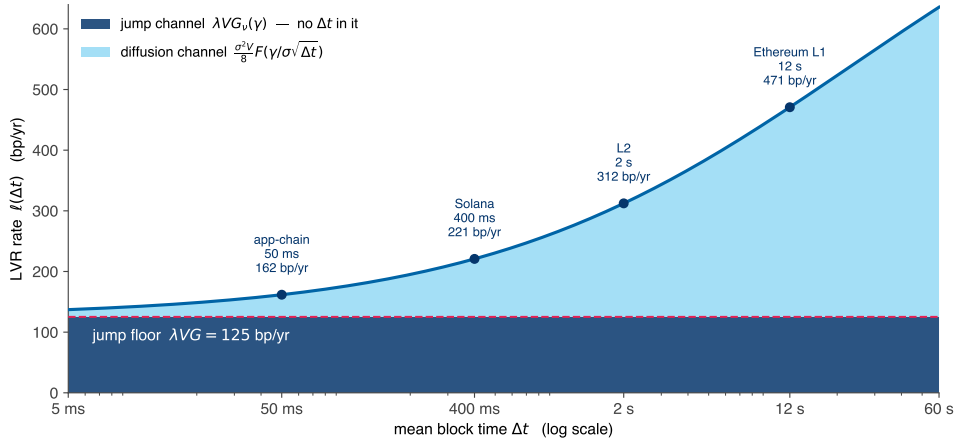
$$\ell(\Delta t) \geq \lambda V G_\nu(\gamma) = \inf_{\Delta t' > 0} \ell_0(\Delta t') > 0.$$

- Proved from convexity of the quadratic surrogate — it never touches the stationary law, which for $\lambda > 0$ has no closed form. That is what makes it available, and why it does *not* inherit Proposition 1's mixing condition.
- **Corollary 1.** ℓ_0 is strictly increasing and the floor is approached only as $\sqrt{\Delta t}$: halving the block time removes at most 29% of the diffusion term.

Keep two limits apart

$\Delta t \rightarrow 0$ shortens the interval between blocks. The jump law is exogenous to the schedule, so it does *not* shrink J .

The two channels, drawn



$\ell_0(\Delta t)$ at the ETH/USDT calibration. The dark band is constant in Δt ; only the pale band responds to the block schedule.

Fees split LVR — they do not reduce it

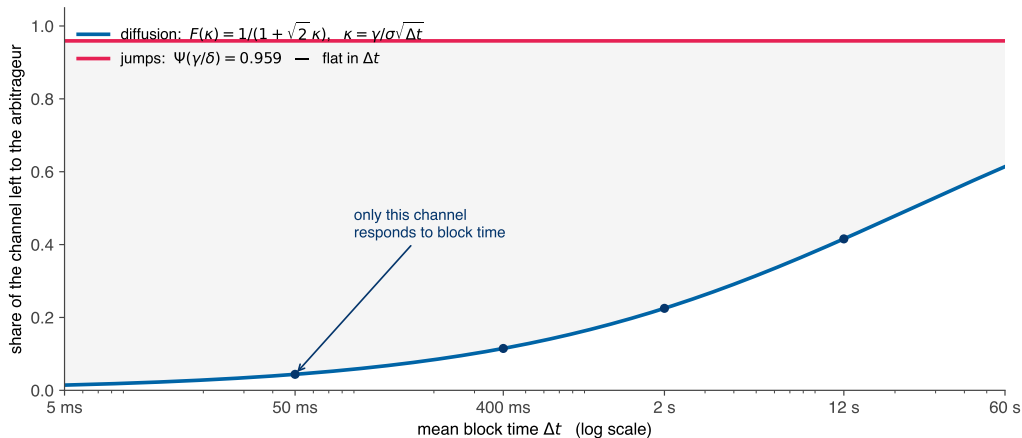
In the accounting of Milionis et al., arbitrage profit and fee income sum to the frictionless LVR of Theorem 1. So $\ell(\Delta t)$ is the arbitrage half: LP loss *net* of the fees LPs get back.

- Diffusion channel: the split is $F(\kappa) : 1 - F(\kappa)$, and $F(\kappa) \rightarrow 0$. Shorter blocks *do* return diffusion-LVR to LPs as fee income.
- Jump channel: the split is $\Psi(\gamma/\delta) : 1 - \Psi(\gamma/\delta)$ — a function of γ/δ alone, with no Δt in it.
- At the calibration $\gamma/\delta = 0.026$, so fees recover 4.1% of jump-LVR. The rest is the floor.

Key takeaway

The lever against the floor is the ratio of fee to jump size — fee-tier design — not the block schedule.

Two fee discounts, one lever



The share of each channel left to the arbitrageur. F collapses as blocks shorten; $\Psi(\gamma/\delta)$ does not move.

Calibration and magnitudes

Calibration — Binance ETH/USDT, Jan 2020 – Jun 2026

2,373 days, 682,946 five-minute bars.

	value	estimator
σ	0.8156	bipower variation
λ	283.3/yr	Lee–Mykland, $\alpha = 1\%$
δ	0.0192	detected jumps, bias-corrected
m	0	$\hat{m} = -0.0012$; 99% CI $\ni 0$
γ	5 bp	Uniswap 0.05% tier
V	\$1M	pool TVL
c	$\$260 \times 10^{-5}$	per block, issuance-derived

Two anchors

- diffusion ceiling
 $\sigma^2 V/8 = 832 \text{ bp/yr}$ (as $\Delta t \rightarrow \infty$)
- jump floor $\lambda VG = 125 \text{ bp/yr}$ (as $\Delta t \rightarrow 0$)

c : Ethereum gross issuance ($\sim \$1.7\text{B/yr}$), scaled for staking above the security-optimal 1/4, attributed pro-rata by value secured.

The rate across deployed block times

Chain	Δt	ℓ (bp/yr)	diff. share
Ethereum L1	12 s	471	73%
Base / OP L2	2 s	312	60%
Solana	400 ms	221	43%
Arbitrum	250 ms	203	38%
App-chain	50 ms	162	23%
Jump floor	—	125	0%

- A $240\times$ cut, $12\text{ s} \rightarrow 50\text{ ms}$, removes only 89% of the diffusion term and leaves the rate 29% above the floor.
- The channels are equal at $\Delta t_{\times} \approx 0.75\text{ s}$.
- The diffusion term reaches a tenth of the floor only at $\Delta t \approx 5.5\text{ ms}$.

The reset-to-zero shortcut

Using $\Psi(\kappa)$ in place of P_{trade} — i.e. assuming a trade resets z to 0 rather than to the band edge — understates the diffusion term by $2.7\times$ at 12 s, $76\times$ at 2 s, and 4×10^7 at 400 ms.

Which regime a chain is in matters more than its slot

ℓ in bp/yr at $\Delta t = 12$ s, diffusion share in brackets:

$\sigma \setminus \lambda$	71/yr	283/yr	1,133/yr	4,533/yr
0.30 (calm)	55 (43%)	148 (16%)	524 (4%)	2,027 (1%)
0.816 (baseline)	377 (92%)	471 (73%)	847 (41%)	2,349 (15%)
1.50 (stress)	1,625 (98%)	1,719 (93%)	2,095 (76%)	3,597 (44%)

- The *level* spans a factor of 65; the *share* runs from 1% to 98% and moves the *other* way in λ .
- The share is what decides whether shortening blocks helps at all. At $\lambda = 283/\text{yr}$ it crosses 50% at $\sigma = 0.55$.

Key takeaway

Any policy claim about 12-second blocks is conditional on the volatility-and-jump-intensity regime, not on a single point estimate.

The planner's block-time optimum

Theorem 4 — netting LVR against what a block costs

Fixed per-block work (gossip, BLS aggregation, view changes) gives a cost rate $R(\Delta t) = r/\Delta t$; a pool inherits $c/\Delta t$ with $c = r V/V_{\text{chain}}$. The LP-side planner minimises $W(\Delta t) = \ell_0(\Delta t) + c/\Delta t$:

$$\eta^{\text{opt}}(1 + \eta^{\text{opt}})^2 = \frac{V\gamma^2}{8c}, \quad \Delta t^{\text{opt}} = \frac{2\gamma^2}{\sigma^2(\eta^{\text{opt}})^2},$$

a cubic with a unique positive root, closed form by Cardano.

Four sensitivities:

- $\partial\Delta t^{\text{opt}}/\partial\lambda = \partial\Delta t^{\text{opt}}/\partial m = \partial\Delta t^{\text{opt}}/\partial\delta = 0$ — **the jump parameters do not enter the first-order condition.**
- $\partial\Delta t^{\text{opt}}/\partial V = 0$, exactly: V cancels, since $c \propto V$. The optimum is a property of the chain and the fee tier, not of the pool.
- $\partial\Delta t^{\text{opt}}/\partial c > 0$; and $\partial\Delta t^{\text{opt}}/\partial\gamma$ is U-shaped.

How far does that invariance actually go?

Honest bookkeeping

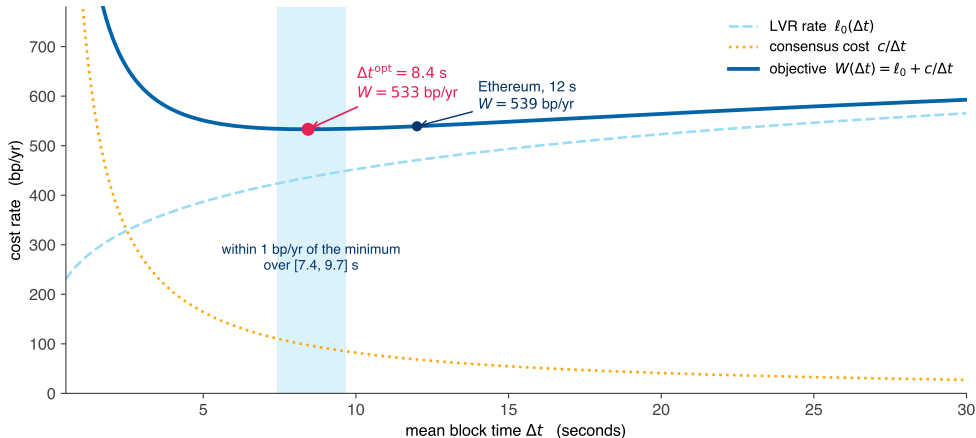
The invariance in (λ, m, δ) is a property of ℓ_0 , whose jump term is an additive constant in Δt . In the *exact* model $\mathcal{E}$ carries both λ and Δt , so the exact first-order condition is not jump-free. It is not a theorem about ℓ .

Bound it directly instead. Any minimiser of $\ell + c/\Delta t$ lies in $\{\Delta t : W(\Delta t) + \mathcal{E}^-(\Delta t) \leq \min_s [W(s) + \mathcal{E}^+(s)]\}$, which at the calibration is [7.3, 9.9] s around the separated optimum $\Delta t^{\text{opt}} = 8.4$ s.

Key takeaway

That is also the range over which W sits within 1 bp/yr of its minimum anyway. So: jump parameters move Δt^{opt} by less than the flatness of the objective can resolve — quantitative, not absolute.

The objective and its optimum



$W(\Delta t) = \ell_0(\Delta t) + c/\Delta t$ at $c = \$260 \times 10^{-5}/\text{block}$. The optimum is shallow: $539 \rightarrow 533 \text{ bp/yr}$, a saving under 2%.

What actually moves the optimum

$c (\times 10^{-5} \text{ USD/block})$	Δt^{opt}
500	15.4 s
260 (baseline)	8.4 s
90	3.4 s
30	1.4 s
2	0.20 s

σ	Δt^{opt}
0.30 (calm)	62.3 s
0.816	8.4 s
1.50 (stress)	2.5 s

- Nearly two orders of magnitude across the plausible range of the cost calibration; volatility moves it comparably, since $\Delta t^{\text{opt}} \propto \gamma^2 / \sigma^2$ at fixed η .
- The jump parameters do not move it at all.

Key takeaway

Under an LP-LVR-only objective, Ethereum's 12 s is somewhat too slow and sub-second targets overshoot — but the gain over the status quo is under 2%, and the answer is driven by c , σ , γ .

What this means, and what it does not

Policy

- Shortening blocks can at most close $\ell(\Delta t) - \lambda VG$, and because F decays algebraically that gap closes only as $\sqrt{\Delta t}$.
- At the calibration the diffusion channel stays material into the low-millisecond range. **So at every deployed block time, what limits block-time policy is not the floor — it is the cost of consensus.**
- Once the floor does bind, the remaining LP-side levers are not the schedule:
 - ▶ fee-tier design, since the binding ratio is γ/δ ;
 - ▶ batch auctions, which remove the per-block arbitrage window at source;
 - ▶ LVR-rebate designs, e.g. auction-managed AMMs routing pool-management revenue back to LPs.

Key takeaway

Jumps change what block time can *achieve*, not what it should *be*.

Limitations

- **Objective is LP-LVR-only.** A full welfare treatment must also weigh MEV redistribution, finality, censorship resistance and oracle freshness — all of which could push Δt^{opt} back up. This is an input, not a prescription.
- **Consensus cost.** Issuance is a chosen compensation policy, not a measured resource cost; it overstates where it subsidises growth and understates where MEV and fees fund validators.
- **Full-range pool.** Most LVR is paid on concentrated liquidity, where the two effects on G run in opposite directions — so the full-range floor is not simply a lower bound there.
- **Idealised arbitrageur:** unlimited capital, zero latency, continuous sight of S_t — weakest exactly in the sub-second regime the floor argument concerns.
- **Poisson blocks**, inherited for tractability. Deterministic slots give the same shape with $C = -\zeta(1/2)/\sqrt{2\pi} \approx 0.583$ in place of $1/\sqrt{2}$, and move Δt^{opt} by 5%.

Conclusion

125

bp/yr jump floor,
exact

471

bp/yr at
Ethereum's 12 s

8.4 s

LP-side
optimum

Key takeaway

The rate splits into a channel the block schedule governs and one it cannot touch. The floor is genuine but distant — what binds at deployed block times is the cost of a block.

Extended version, with all proofs: [arXiv:2608.30321](https://arxiv.org/abs/2608.30321) ↗.

Questions and discussion.

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